

RIO GRANDE, Argentina

WMO: 879340

Lat: 53.7803S

Lon: 67.7586W

Elev: 72

StdP: 14.66

Time Zone: -3.00 (W03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	18.7	21.3	14.4	11.5	18.8	18.0	13.6	22.2	32.7	40.4	30.1	39.5	5.6	230	0.926

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	67.6	58.5	64.3	55.5	62.0	54.1	59.1	65.9	56.7	62.8	54.6	60.5	25.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.6	65.9	63.0	53.3	60.6	60.1	51.4	56.5	57.6	25.8	66.0	24.3	63.2	23.0	60.5	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.1	33.2	30.3	DB	11.7	74.2	4.3	3.0	8.6	76.4	6.0	78.1	3.6	79.8	0.4	82.0	(4)
(5)			WB	10.8	64.3	5.0	2.4	7.2	66.0	4.3	67.4	1.5	68.7	-2.1	70.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	51.5	51.3	47.3	42.7	37.5	33.3	33.8	35.8	39.5	43.2	46.6	50.2	(6)
(7)		DBStd	7.83	3.97	5.15	4.50	4.46	5.17	5.90	4.54	3.76	4.08	3.95	4.17	3.53	(7)
(8)		HDD50	2915	28	40	109	220	388	501	502	440	315	214	116	42	(8)
(9)		HDD65	8147	417	384	550	668	852	951	967	905	765	677	551	460	(9)
(10)		CDD50	243	76	76	24	2	0	0	0	0	0	3	15	47	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	10	5	5	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	15.1	19.2	17.2	15.8	14.5	11.6	10.7	11.5	13.2	13.7	16.8	18.5	19.1	(14)
(15)	Precipitation	PrecAvg	12.8	1.3	0.9	1.1	1.4	1.3	1.4	0.9	0.7	0.6	1.1	1.3	(15)	
(16)		PrecMax	19.1	2.7	2.5	3.0	2.8	3.0	4.3	2.4	1.7	1.7	1.5	2.5	2.9	(16)
(17)		PrecMin	8.4	0.5	0.2	0.3	0.1	0.2	0.5	0.2	0.0	0.1	0.0	0.2	0.3	(17)
(18)		PrecStd	3.0	0.7	0.6	0.6	0.7	0.7	1.0	0.7	0.5	0.5	0.4	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	74.9	66.4	59.0	53.7	49.9	46.6	49.8	57.0	61.0	64.8	67.7	(19)
(20)			MCWB	59.3	63.0	58.0	52.5	49.3	47.4	43.6	44.9	49.6	53.1	55.6	56.7	(20)
(21)		2%	DB	66.1	68.3	61.0	55.6	50.1	46.1	43.2	46.2	52.2	57.4	61.1	64.3	(21)
(22)			MCWB	56.7	60.2	54.1	50.2	46.6	44.0	41.1	42.0	45.9	49.0	52.8	54.2	(22)
(23)		5%	DB	62.8	64.5	58.8	53.6	47.8	43.1	41.3	43.2	49.8	55.2	58.8	61.2	(23)
(24)			MCWB	54.7	56.6	52.2	48.7	45.1	41.9	39.7	40.2	44.0	47.5	50.6	52.4	(24)
(25)		10%	DB	60.6	61.1	55.8	51.5	46.0	41.2	40.6	41.4	46.8	52.1	55.7	59.1	(25)
(26)			MCWB	52.9	54.3	50.4	47.7	44.0	39.9	39.1	39.2	42.3	45.5	48.7	51.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	64.3	59.2	53.3	50.2	47.3	43.7	45.0	50.2	53.1	57.2	57.6	(27)
(28)			MCD B	68.3	72.1	65.5	57.3	52.6	48.8	45.4	47.9	56.1	60.5	64.6	65.4	(28)
(29)		2%	WB	57.7	60.7	55.2	51.3	47.0	43.8	41.4	42.3	46.7	50.1	53.4	55.4	(29)
(30)			MCD B	64.6	67.2	59.8	54.8	48.9	44.9	42.4	44.5	51.2	56.0	60.4	62.3	(30)
(31)		5%	WB	55.5	58.0	52.8	49.8	45.3	41.9	40.1	40.7	44.3	48.0	51.1	53.2	(31)
(32)			MCD B	61.5	63.8	57.3	52.4	46.7	42.9	41.0	42.4	48.1	53.8	57.1	59.9	(32)
(33)		10%	WB	53.4	55.0	51.2	48.0	43.6	39.9	38.7	39.5	42.6	46.1	49.6	51.7	(33)
(34)			MCD B	59.6	60.3	54.9	50.7	45.1	40.9	39.7	41.1	45.9	51.1	55.3	57.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.2	17.3	16.3	15.3	13.3	10.5	10.1	11.6	14.3	16.5	16.8	17.4	(35)
(36)			MCDBR	19.4	21.6	18.7	17.1	14.7	10.9	10.2	13.5	17.3	20.0	19.9	20.4	(36)
(37)			MCWBR	13.0	14.5	13.5	13.2	12.1	9.6	8.8	10.6	12.3	13.4	13.3	13.3	(37)
(38)		5% WB	MCD BR	18.8	20.5	17.3	16.0	13.1	10.4	9.4	12.4	15.9	18.8	19.2	19.6	(38)
(39)			MCWBR	14.0	14.7	13.5	13.2	11.6	9.6	8.5	10.3	12.1	13.5	13.7	13.6	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.319	0.313	0.312	0.302	0.285	0.270	0.281	0.299	0.309	0.314	0.319	0.322	(41)	
(42)		taud	2.553	2.575	2.562	2.550	2.457	2.364	2.420	2.491	2.520	2.516	2.514	2.523	(42)	
(43)		Ebn at Noon	312	302	282	251	212	189	209	246	278	298	310	314	(43)	
		Edh at Noon	31	29	25	20	15	13	15	20	25	30	32	33		
(44)	All-Sky Solar Radiation	RadAvg	1819	1511	1000	572	292	184	229	457	864	1381	1771	1948	(44)	
(45)		RadStd	90	73	72	32	23	11	16	31	58	55	89	114	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page